

Genera of Curves: Algebraic Geometry and Polytopes

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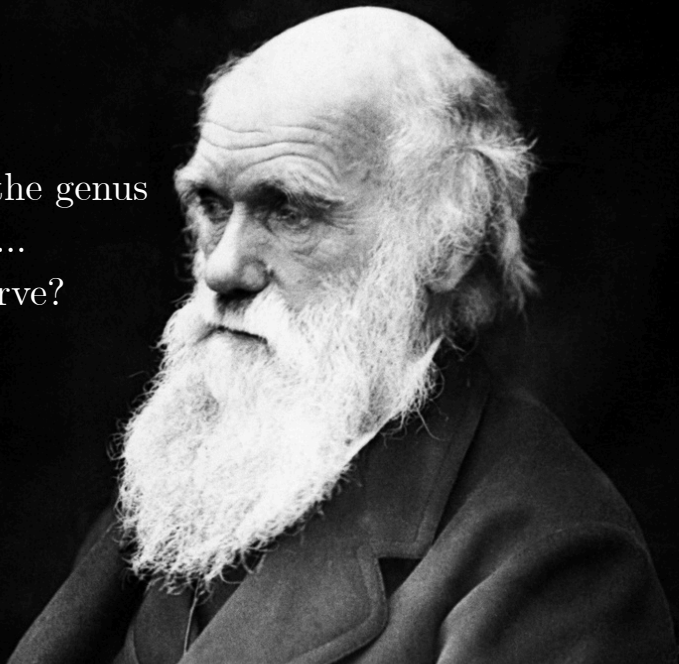
Genus? Genera?

- What is the genus of a curve?
- Can a curve have **high genus** but still be defined by a **simple equation**?
- What controls the genus: degree, or something deeper?

Goal

- Understand genus through increasingly rich geometric and combinatorial perspectives.

What is the genus
of...
a curve?



Genus for you geniuses

- Polynomial $f(x, y)$ and its vanishing locus $V(f)$

Genus = number of “holes” of the Riemann surface of the curve over $V(f)$

Encountering genus for smooth curves

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Key idea:

- The genus governs the complexity of functions and differentials on C .

Genus of smooth curves

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$$g = \dim\{\text{holomorphic differentials on } C\}$$

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Interpretation:

- Topological: number of “holes”
- Algebraic: number of independent differential forms on C

Example

- Consider projective curves of degree d :

$$f(x, y) = 0$$

- If smooth:

$$g = \frac{(d-1)(d-2)}{2}$$

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Examples:

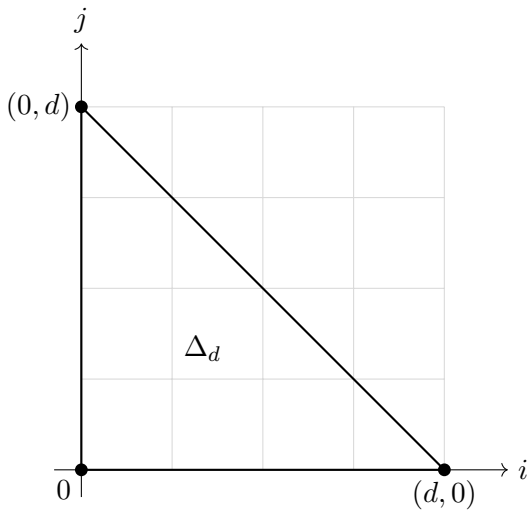
- Line ($d = 1$): $g = 0$
- Conic ($d = 2$): $g = 0$
- Cubic ($d = 3$): $g = 1$

The equation:

$$g = \frac{(d-1)(d-2)}{2} = \binom{d-1}{2}$$

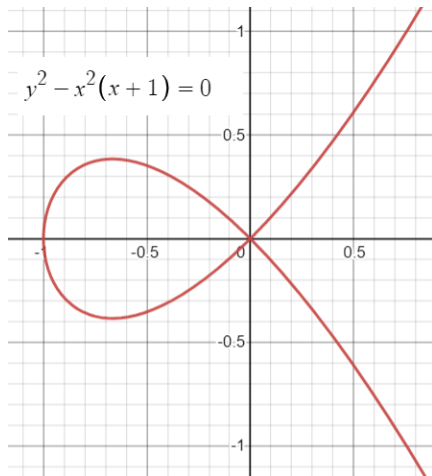
... is the number of interior points of the triangle with length d

Geometric picture



A non-example

- Singular curves break the naive formula
- Example: nodal cubic



What about non-smooth curves?

Problem:

- Smooth genus formula no longer applies directly
- Need a new framework

Resolution of Singularities

Hironaka's Theorem:

- Every algebraic variety admits a resolution of singularities
- Achieved via a sequence of **blow-ups**

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*“Every blowup simplifies the singularity but complicates
your life.”*

— Prof. John Voight

Nondegenerate curves

- Consider curves defined by Laurent polynomials:

$$f(x, y) \in k[x^{\pm 1}, y^{\pm 1}] = k[\mathbb{Z}^2]$$

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- Associated **Newton polytope**:

Definition (Newton Polytope)

Convex hull of exponents in $\mathbb{R}^2$

Example:

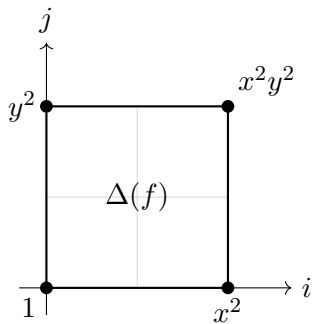
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$$x^2y^2 \mapsto (2, 2), x^2 \mapsto (2, 0), y^2 \mapsto (0, 2), 1 = x^0y^0 \mapsto (0, 0)$$

Example:



Nondegeneracy

Definition

Let $f|_\tau$ be the polynomial $f|_\tau = \sum_{(i,j) \in \tau} c_{ij} x^i y^j$.

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Definition (Nondegeneracy)

$$\begin{cases} f|_\tau = 0 \\ x \frac{\partial f|_\tau}{\partial x} = 0 \\ y \frac{\partial f|_\tau}{\partial y} = 0 \end{cases} \quad \text{has no solutions in } (k^*)^2.$$

Nondegeneracy

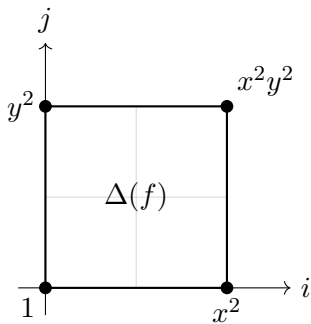
Theorem

f is nondegenerate $\implies$ The closure $\overline{V(f)} \subseteq X_\Gamma$ is smooth.

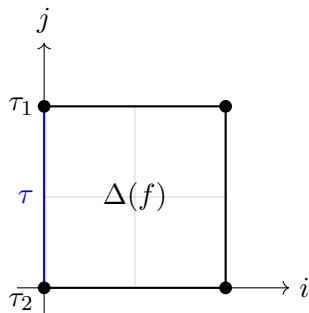
Analogy



Constructing X_Γ



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From faces to geometry

- Each face $\tau \subset \Delta(f)$ determines algebraic data
- Associate an affine toric variety:

$$U_\tau = \operatorname{Spec} k[S_\tau]$$

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Key idea:

- The Newton polytope encodes a *toric surface*

Face inclusions $\leftrightarrow$ open embeddings

- If $\tau_1 \subset \tau$, then

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Interpretation:

- Smaller faces $\Rightarrow$ bigger affine pieces
- Overlaps come from shared subfaces

Gluing the toric surface

- Glue all U_τ along these inclusions
- Produces a toric variety X_Γ

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Analogy:

- Like gluing coordinate charts on a manifold

Toric compactification

- The torus $(k^*)^2$ sits inside X_Γ
- Boundary = union of divisors from edges

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Key idea:

- Edges correspond to boundary components

Example: The projective plane

Take the triangle:

$$\Delta = \text{Conv}\{(0,0), (1,0), (0,1)\}$$

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Interpretation:

- Gluing toric charts recovers a familiar space

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Conclusion:

- Geometry of $V(f)$ controlled by faces of Δ

Baker's Theorem

Theorem (Baker, 1893)

Let $f(x, y)$ in $k[x, y]$ be nondegenerate. Then the closure $V(f)$ in X_Γ is a smooth projective plane curve of genus equal to the number of interior lattice points in Δ .

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Remark:

- Genus becomes a **combinatorial quantity**

Example: Applying Baker's Theorem

Consider the Laurent polynomial

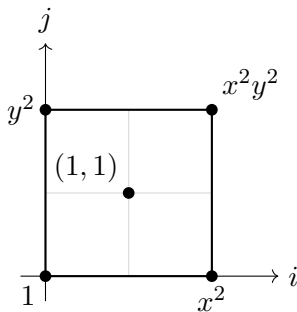
$$f(x, y) = 1 + x^2 + y^2 + x^2y^2.$$

Example: Applying Baker's Theorem

Consider the Laurent polynomial

$$f(x, y) = 1 + x^2 + y^2 + x^2y^2.$$

Its Newton polytope is the convex hull of the exponent vectors.



Example: Applying Baker's Theorem

Baker's Theorem:

$$g(C) = \#(\text{Int}(\Delta(f)) \cap \mathbb{Z}^2)$$

Here there is exactly one interior lattice point, so:

$$g(C) = 1.$$

Classification by genus

Theorem (Castryck-Voight)

Every smooth projective curve C of genus $g \leq 4$ is nondegenerate, i.e., $C = V(f)$ for f in $k[x, y]$ nondegenerate. Moreover, the generic curve of genus $g \geq 5$ is not nondegenerate.

Target: genus $g = 5$

- We now reverse the question:

Can we *construct* curves of a given genus?

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Two approaches:

- Classical: complete intersections
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Two approaches:

- Classical: complete intersections
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Goal: make these two viewpoints meet

Complete intersections

- Curves as intersections:

$$C = V(f_1, f_2) \subset \mathbb{P}^3$$

- Geometry is tightly controlled

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Key idea:

- Genus can be computed from defining equations
- We can aim for $g = 5$ by choosing equations carefully

Combinatorial viewpoint

- Each polynomial has a Newton polytope:

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- Their interaction controls the geometry

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Theorem (Khovanskii, 1978)

The genus of a complete intersection $X = V(f_1, f_2)$ is given by

$$g(X) = \#(\text{Int}(\Gamma_1 + \Gamma_2)) - \#(\text{Int}(\Gamma_1)) - \#(\text{Int}(\Gamma_2))$$

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Interpretation:

- Genus = interior lattice points gained by combining polytopes

Strategy

- Choose polytopes $\Gamma_1, \Gamma_2 \subset \mathbb{Z}^3$
- Compute:

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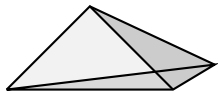
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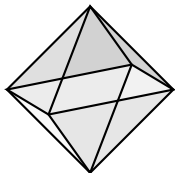
Plan:

- Try examples
- Adjust combinatorics until genus matches

Comparing polytopes



Γ_1



Γ_2

A successful construction

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- This pair realizes a genus 5 curve
- There is a class of such pairs

Conclusion

- Genus appears in multiple forms:
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- Genus appears in multiple forms:
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- Newton polytopes translate geometry into counting
- We can *construct* curves of fixed genus via combinatorics

Geometry $\longleftrightarrow$ Combinatorics